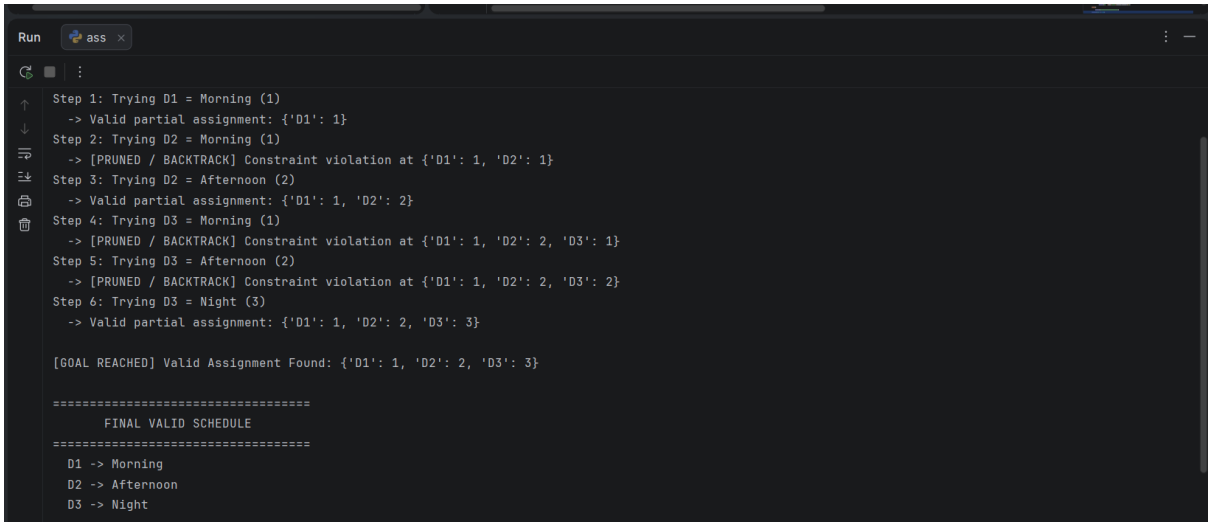
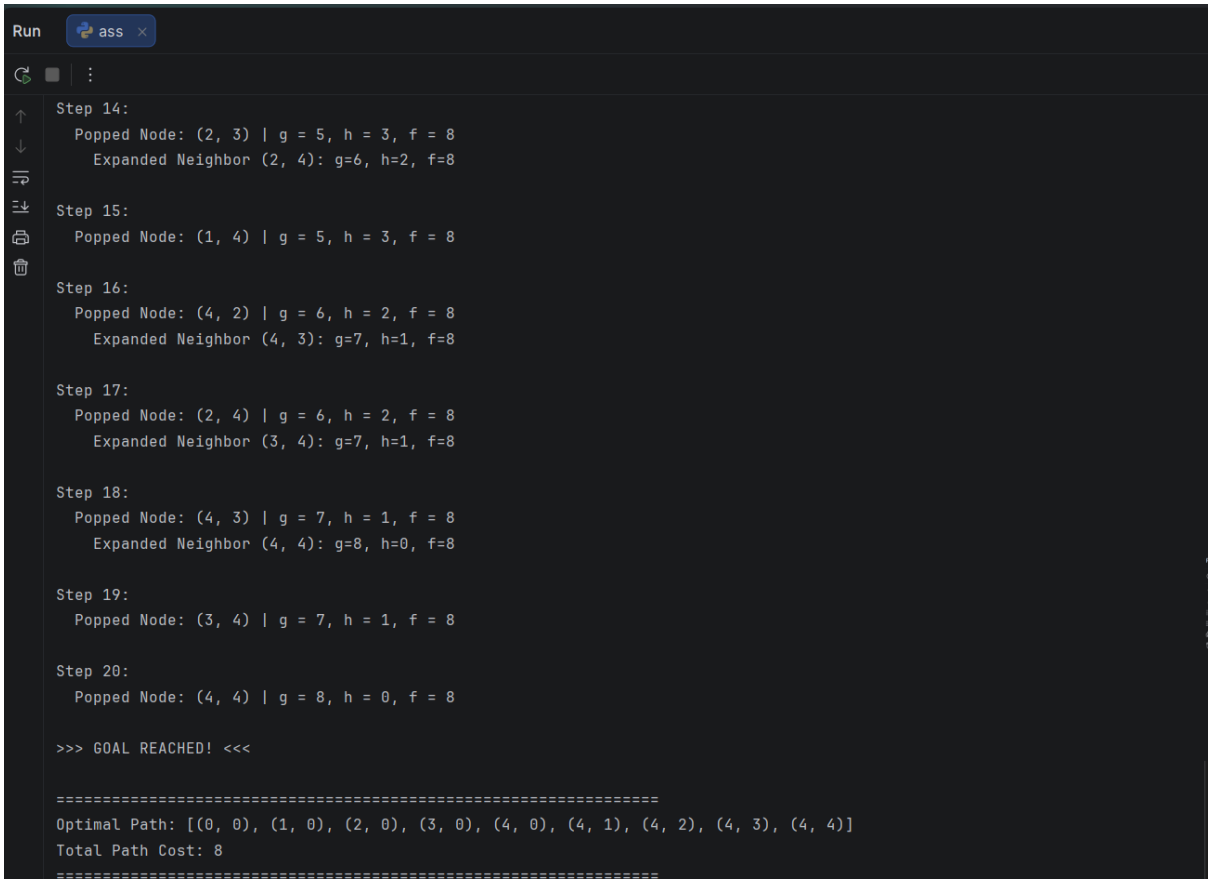
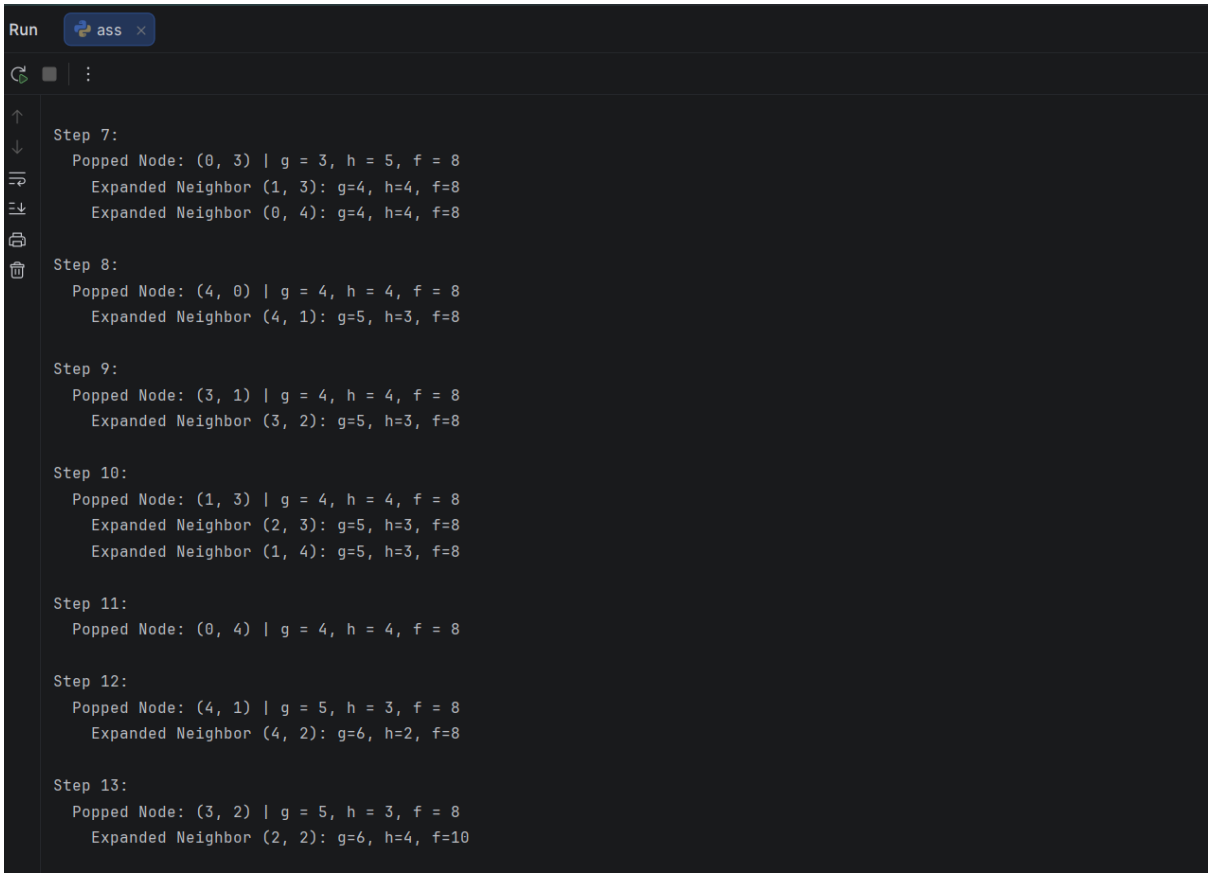
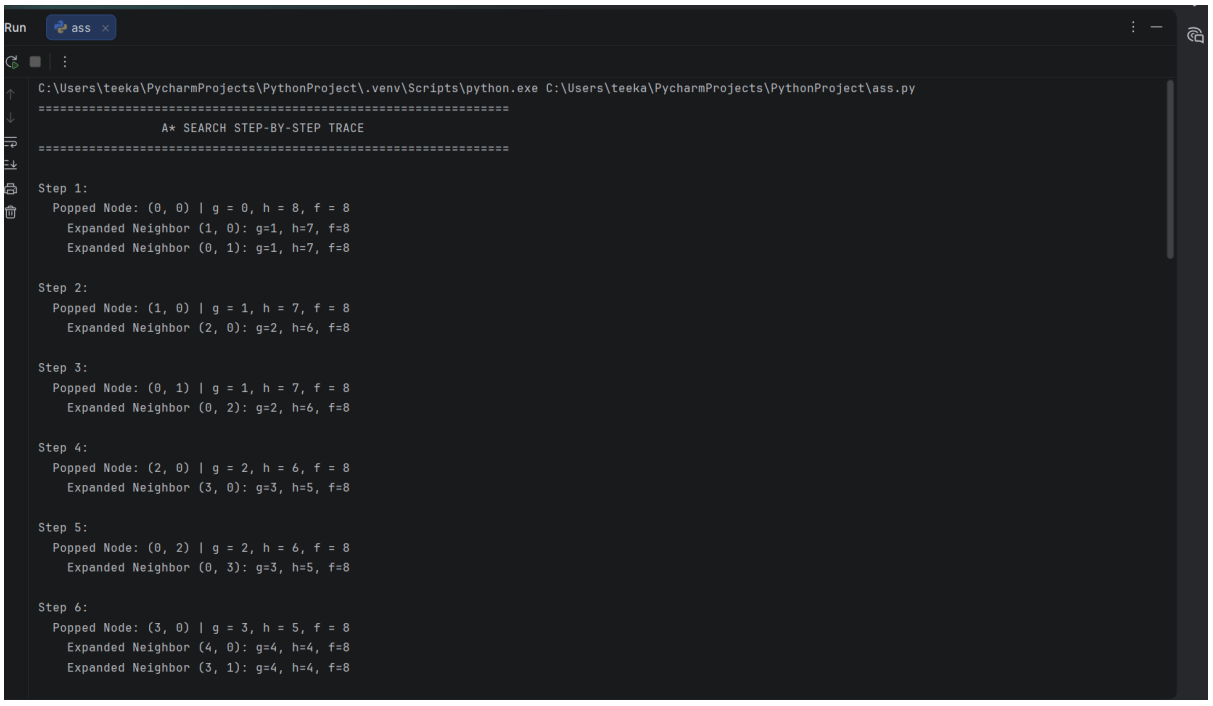
1)


Run
ass
Step 1: Trying D1 = Morning (1)
-> Valid partial assignment: {'D1': 1}
Step 2: Trying D2 = Morning (1)
-> [PRUNED / BACKTRACK] Constraint violation at {'D1': 1, 'D2': 1}
Step 3: Trying D2 = Afternoon (2)
-> Valid partial assignment: {'D1': 1, 'D2': 2}
Step 4: Trying D3 = Morning (1)
-> [PRUNED / BACKTRACK] Constraint violation at {'D1': 1, 'D2': 2, 'D3': 1}
Step 5: Trying D3 = Afternoon (2)
-> [PRUNED / BACKTRACK] Constraint violation at {'D1': 1, 'D2': 2, 'D3': 2}
Step 6: Trying D3 = Night (3)
-> Valid partial assignment: {'D1': 1, 'D2': 2, 'D3': 3}
[GOAL REACHED] Valid Assignment Found: {'D1': 1, 'D2': 2, 'D3': 3}
==========
FINAL VALID SCHEDULE
==========
D1 -> Morning
D2 -> Afternoon
D3 -> Night

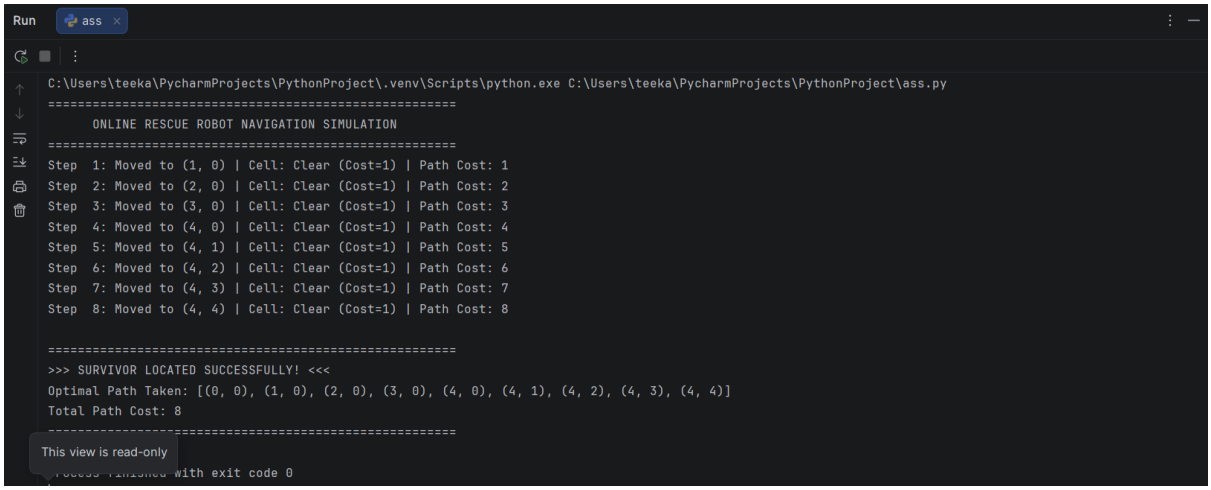
2)


Run
ass
Step 14:
Popped Node: (2, 3) | g = 5, h = 3, f = 8
Expanded Neighbor (2, 4): g=6, h=2, f=8
Step 15:
Popped Node: (1, 4) | g = 5, h = 3, f = 8
Step 16:
Popped Node: (4, 2) | g = 6, h = 2, f = 8
Expanded Neighbor (4, 3): g=7, h=1, f=8
Step 17:
Popped Node: (2, 4) | g = 6, h = 2, f = 8
Expanded Neighbor (3, 4): g=7, h=1, f=8
Step 18:
Popped Node: (4, 3) | g = 7, h = 1, f = 8
Expanded Neighbor (4, 4): g=8, h=0, f=8
Step 19:
Popped Node: (3, 4) | g = 7, h = 1, f = 8
Step 20:
Popped Node: (4, 4) | g = 8, h = 0, f = 8
>>> GOAL REACHED! <<<
==========
Optimal Path: [(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
Total Path Cost: 8
==========


Run
ass
Step 7:
Popped Node: (0, 3) | g = 3, h = 5, f = 8
Expanded Neighbor (1, 3): g=4, h=4, f=8
Expanded Neighbor (0, 4): g=4, h=4, f=8
Step 8:
Popped Node: (4, 0) | g = 4, h = 4, f = 8
Expanded Neighbor (4, 1): g=5, h=3, f=8
Step 9:
Popped Node: (3, 1) | g = 4, h = 4, f = 8
Expanded Neighbor (3, 2): g=5, h=3, f=8
Step 10:
Popped Node: (1, 3) | g = 4, h = 4, f = 8
Expanded Neighbor (2, 3): g=5, h=3, f=8
Expanded Neighbor (1, 4): g=5, h=3, f=8
Step 11:
Popped Node: (0, 4) | g = 4, h = 4, f = 8
Step 12:
Popped Node: (4, 1) | g = 5, h = 3, f = 8
Expanded Neighbor (4, 2): g=6, h=2, f=8
Step 13:
Popped Node: (3, 2) | g = 5, h = 3, f = 8
Expanded Neighbor (2, 2): g=6, h=4, f=10


Run
ass
C:\Users\teeka\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\teeka\PycharmProjects\PythonProject\ass.py
A* SEARCH STEP-BY-STEP TRACE
Step 1:
Popped Node: (0, 0) | g = 0, h = 8, f = 8
Expanded Neighbor (1, 0): g=1, h=7, f=8
Expanded Neighbor (0, 1): g=1, h=7, f=8
Step 2:
Popped Node: (1, 0) | g = 1, h = 7, f = 8
Expanded Neighbor (2, 0): g=2, h=6, f=8
Step 3:
Popped Node: (0, 1) | g = 1, h = 7, f = 8
Expanded Neighbor (0, 2): g=2, h=6, f=8
Step 4:
Popped Node: (2, 0) | g = 2, h = 6, f = 8
Expanded Neighbor (3, 0): g=3, h=5, f=8
Step 5:
Popped Node: (0, 2) | g = 2, h = 6, f = 8
Expanded Neighbor (0, 3): g=3, h=5, f=8
Step 6:
Popped Node: (3, 0) | g = 3, h = 5, f = 8
Expanded Neighbor (4, 0): g=4, h=4, f=8
Expanded Neighbor (3, 1): g=4, h=4, f=8

3)


Run
ass
C:\Users\teeka\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\teeka\PycharmProjects\PythonProject\ass.py
ONLINE RESCUE ROBOT NAVIGATION SIMULATION
Step 1: Moved to (1, 0) | Cell: Clear (Cost=1) | Path Cost: 1
Step 2: Moved to (2, 0) | Cell: Clear (Cost=1) | Path Cost: 2
Step 3: Moved to (3, 0) | Cell: Clear (Cost=1) | Path Cost: 3
Step 4: Moved to (4, 0) | Cell: Clear (Cost=1) | Path Cost: 4
Step 5: Moved to (4, 1) | Cell: Clear (Cost=1) | Path Cost: 5
Step 6: Moved to (4, 2) | Cell: Clear (Cost=1) | Path Cost: 6
Step 7: Moved to (4, 3) | Cell: Clear (Cost=1) | Path Cost: 7
Step 8: Moved to (4, 4) | Cell: Clear (Cost=1) | Path Cost: 8
>>> SURVIVOR LOCATED SUCCESSFULLY! <<<
Optimal Path Taken: [(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
Total Path Cost: 8
This view is read-only
Process finished with exit code 0